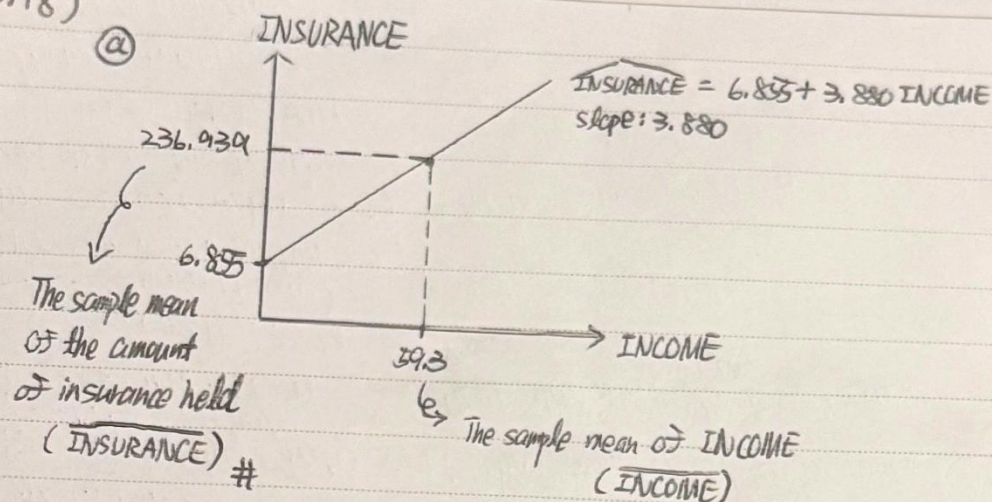


(3.18)

(3.18)



- ⑥ We estimate that for each additional \$1000 of household income, the average amount of insurance held increases by \$3,880. #

point estimate: 3.880 #

95% CI for B_2 : $b_2 \pm t_{0.025}(18) se(b_2) = 3.880 \pm 2.101 \times 0.112$
 $\Rightarrow [3.6447, 4.1153] \#$

We have 95% confidence that the actual increase in insurance demand for every \$1000 of income growth lies within the range \$3,645 to \$4,115. #

⑦ $E(\widehat{INSURANCE} | INCOME = 100) = 6.855 + 3.880 \times 100 = 394.855$

Let $INSURANCE = Y$, $INCOME = X \rightarrow Y = B_1 + B_2 X_0$

$Var(\hat{Y}) = Var(b_1 + b_2 X_0) = Var(b_1) + X_0^2 Var(b_2) + 2X_0 Cov(b_1, b_2) = 7.383^2 + 100^2 \times 0.112^2 + 2 \times 100 \times (-0.746) = 30.7487$

99% C.I. for $E(\widehat{INSURANCE} | INCOME = 100)$: $394.855 \pm t_{0.005}(18) \times \sqrt{30.7487}$
 $\Rightarrow [378.897, 410.813] \#$

d)

$$1. \begin{cases} H_0: \beta_2 = 5 \\ H_a: \beta_2 \neq 5 \end{cases}$$

2. $\alpha = 0.05$

3. Test statistic: $t = \frac{b_2 - 5}{0.112} \sim t(18)$

4. $t^* = \frac{3.880 - 5}{0.112} = -10$

5. $RR = \{t: |t| > t_{0.025}(18) = 2.101\}$

6. $t^* = -10 \in RR$, reject H_0

The data don't support the claim.

Statistically, there is no significant evidence that β_2 equal 5. #

e)

$$1. \begin{cases} H_0: \beta_2 = 1 \\ H_a: \beta_2 > 1 \end{cases}$$

2. $\alpha = 0.01$

3. test statistic: $t = \frac{b_2 - 1}{0.112} \sim t(18)$

4. $t^* = \frac{3.880 - 1}{0.112} \approx 25.7143$

5. $RR = \{t: t > t_{0.01}(18) = 2.552\}$

6. $t^* = 25.73 \in RR$, reject H_0

It is statistically significant that as income increases, the amount of life insurance held increases by a larger amount.

(3.23)

(a)

At the 5% significance level, we fail to reject the null hypothesis, because the calculated p-value of 1 is significantly greater than the alpha level of 0.05 (and the t-statistic of -26.729 does not fall within the rejection region). This indicates that there is insufficient statistical evidence to conclude that the marginal effect of increasing house size by 100 square feet for a 2000-square-foot home is greater than \$13,000.

```
> cat("臨界值 (Critical Value):", critical_val, "\n")
臨界值 (Critical Value): 1.647919
> cat("拒絕域 (Rejection Region): t >", critical_val, "\n\n")
拒絕域 (Rejection Region): t > 1.647919

<
> cat("【(a) 小題：2000 sqft】 \n")
【(a) 小題：2000 sqft】
> cat("邊際效應:", me_20, "\nt 統計量:", t_a, "\n右尾 p-value:", p_a, "\n")
邊際效應: 7.38076
t 統計量: -26.72875
右尾 p-value: 1
> cat("結論:", ifelse(t_a > critical_val, "拒絕 H0", "無法拒絕 H0"), "\n\n")
結論: 無法拒絕 H0
```

(b)

At the 5% significance level, we reject the null hypothesis, because the calculated p-value is less than 0.05, providing sufficient statistical evidence to conclude that the marginal effect of increasing house size by 100 square feet for a 4000-square-foot home is significantly greater than \$13,000.

```
> cat("臨界值 (Critical Value):", critical_val, "\n")
臨界值 (Critical Value): 1.647919
> cat("拒絕域 (Rejection Region): t >", critical_val, "\n\n")
拒絕域 (Rejection Region): t > 1.647919

<
> cat("【(b) 小題：4000 sqft】 \n")
【(b) 小題：4000 sqft】
> cat("邊際效應:", me_40, "\nt 統計量:", t_b, "\n右尾 p-value:", p_b, "\n")
邊際效應: 14.76152
t 統計量: 4.189467
右尾 p-value: 1.65395e-05
> cat("結論:", ifelse(t_b > critical_val, "拒絕 H0", "無法拒絕 H0"), "\n\n")
結論: 拒絕 H0
```

(c)

The estimated expected price for a house with 2000 square feet of living area is \$167,373.50. Based on our quadratic regression model, the 95% confidence interval for the mean price of all such houses is between \$158,048.10 and \$176,698.80.

General Audience Description

If we consider all single-family houses in Baton Rouge with exactly 2,000 square feet of living space, we estimate their average selling price to be approximately **\$167,374**. Furthermore, we are 95% confident that the true average market price for houses of this specific size falls within the range of **\$158,048 to \$176,699**. This interval provides a measure of the precision of our estimate, suggesting that while individual house prices may vary due to other factors, the collective average is highly likely to be within this bracket.

```
>
> cat("【(c) 小題：預測區間】\n")
【(c) 小題：預測區間】
> cat("預期價格:", pred_c[1, "fit"], "\n95% CI: [", pred_c[1, "lwr"], ",", pred_c[1, "upr"], "]\n\n")
預期價格: 167.3735
95% CI: [ 158.0481 , 176.6988 ]
```

(d)

The sample average price for houses with exactly 2,000 square feet (SQFT = 20) is \$163.3333. This value falls well within the 95% confidence interval of [158.0481, 176.6988] that we calculated in part (c).

Therefore, the sample average is compatible with the results from the quadratic regression model. This suggests that the model's functional form is a good fit for the data at this specific house size, as the actual observed mean closely aligns with our statistical prediction.

```
>
> cat("【(d) 小題：樣本平均】\n")
【(d) 小題：樣本平均】
> cat("2000 sqft 真實平均房價:", avg_p_20, "\n")
2000 sqft 真實平均房價: 163.3333
> |
```

(3.31)

(a)

The summary statistics for weekly unit sales (SAL1) and price per can (APR1) are as follows:

Unit Sales:

Mean: 6,719 cans

Minimum: 1,762 cans

Maximum: 32,820 cans

Standard Deviation: 6,915.083

Price:

Mean: \$0.7825

Minimum: \$0.6100

Maximum: \$0.9200

Standard Deviation: 0.0980

Variation Analysis:

To compare the variation between the two variables, we look at the **Coefficient of Variation (CV)**:

The CV for sales (SAL1) is **1.0292** (or 102.92%).

The CV for price (APR1) is **0.1253** (or 12.53%).

Conclusion on Variation:

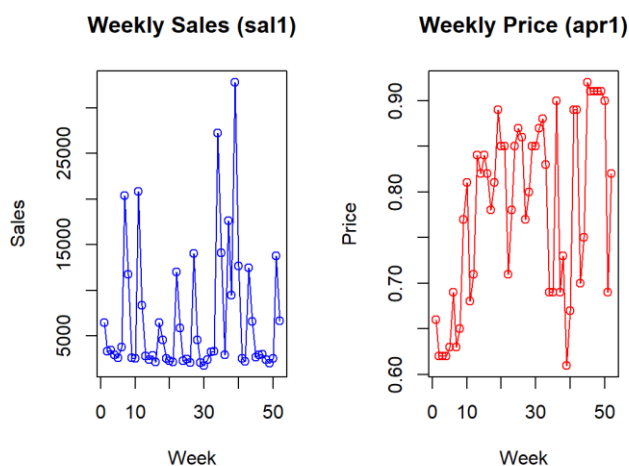
There is **substantially more variation in sales than in price** from week to week. While the price is relatively stable (fluctuating by about 12.5% relative to its mean), the sales volume is highly volatile, with a standard deviation that actually exceeds its mean value ($CV > 1$).

```

> print(summary_stats)
      sal1      apr1
Min.   : 1762   Min.   :0.6100
1st Qu.: 2485   1st Qu.:0.6900
Median : 3136   Median :0.8100
Mean   : 6719   Mean    :0.7825
3rd Qu.: 8612   3rd Qu.:0.8625
Max.   :32820   Max.    :0.9200
> cat("SAL1 標準差:", sd_sal1, "\n")
SAL1 標準差: 6915.083
> cat(" 變異係數 (CV):", round(cv_sal1, 4), "\n\n")
 變異係數 (CV): 1.0292

> cat("APR1 標準差:", sd_apr1, "\n")
APR1 標準差: 0.09803711
> cat(" 變異係數 (CV):", round(cv_apr1, 4), "\n")
 變異係數 (CV): 0.1253
>

```

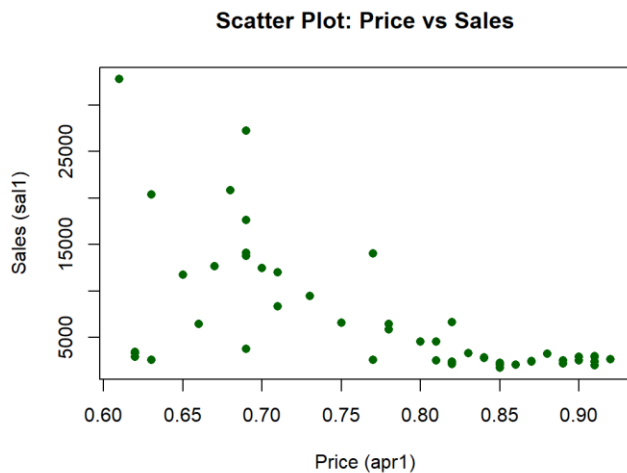


(b)

The scatter plot of weekly unit sales (SAL1) against price (APR1) shows a clear **inverse (negative) relationship**. As the price of the canned tuna increases, the number of units sold tends to decrease.

This inverse relationship is **exactly what we would expect** based on the economic **Law of Demand**, which states that, ceteris paribus (all else being equal), there is an inverse relationship between the price of a good and the quantity demanded.

Consumers are generally price-sensitive when it comes to grocery staples like canned tuna. When the price is high, sales are consistently low. Conversely, the largest spikes in sales occur when the price drops below \$0.70, suggesting that price promotions are a significant driver of consumer purchasing behavior for this brand. The non-linear "clustering" at the lower price range also hints that the relationship might not be perfectly linear, which is common in demand analysis.



(c)

```
> print("--- (c) Regression Results (Price Effect) ---")
[1] "--- (c) Regression Results (Price Effect) ---"
> cat("點估計值 (每增加1美分影響):", b2_hat, "\n")
點估計值 (每增加1美分影響): -434.4473
> cat("95% 信賴區間: [", conf_b2[1], ",", conf_b2[2], "]\n")
95% 信賴區間: [ -592.2897 , -276.6049 ]
> |
```

(d)

```
>
> print("--- (d) Prediction at 70 Cents ---")
[1] "--- (d) Prediction at 70 Cents ---"
> cat("預期銷量 (fit):", pred_d[1, "fit"], "\n")
預期銷量 (fit): 10302.9
> cat("90% 信賴區間: [", pred_d[1, "lwr"], ",", pred_d[1, "upr"], "]\n")
90% 信賴區間: [ 8624.934 , 11980.87 ]
> |
```

(e)

The estimated price elasticity of demand at the sample means is approximately -5.060, with a 95% confidence interval of [-6.898, -3.222]. Since the entire interval is significantly less than -1, the sales of brand no. 1 are highly elastic with respect to price.

This makes strong economic sense because canned tuna is a non-essential grocery item with many close substitutes. Consumers are very price-sensitive; even a small percentage increase in price leads to a more than five-fold percentage decrease in quantity demanded, as they can easily switch to other brands or similar protein sources.

```
> cat("--- (e) Elasticity Results ---\n")
--- (e) Elasticity Results ---
> cat("平均值處的彈性估計值:", elasticity_hat, "\n")
平均值處的彈性估計值: -5.059825
> cat("95% 彈性信賴區間: [", conf_elasticity[1], ",", conf_elasticity[2], "]\n")
95% 彈性信賴區間: [ -6.898149 , -3.221501 ]
> |
```

(f)

At the 10% significance level, we reject the null hypothesis (H_0 : Elasticity = -3) because the p-value of 0.0288 is less than 0.10. There is sufficient statistical evidence to conclude that the price elasticity is significantly different from -3.

```
--- (f) Hypothesis Test ---
> # H0: Elasticity = -3
> # H1: Elasticity != -3 (顯著水準 10%)
> cat("t 統計量:", t_stat, "\n")
t 統計量: -2.250572
> cat("p-value:", p_val, "\n")
p-value: 0.02884204
> cat("10% 顯著水準下的 t 臨界值: +/-", t_crit_10, "\n")
10% 顯著水準下的 t 臨界值: +/- 1.675905
>
> if(p_val < 0.10) {
+   cat("結論：拒絕 H0，彈性顯著不等於 -3\n")
+ } else {
+   cat("結論：不拒絕 H0，沒有足夠證據說明彈性與 -3 有顯著差異\n")
+ }
結論：拒絕 H0，彈性顯著不等於 -3
>
```